2025.2

1) $\cancel{a_1}, \cancel{a_2}, \cancel{a_3}, \cancel{p_2}, \cancel{p_3}, \cancel{q_{21}}, \cancel{q_{31}}, \cancel{q_{32}}$

$\varphi = \frac{F_1+F_2+F_3}{3} \Rightarrow a_1 = a_2 = a_3 = \frac{1}{3}$

$p_2 = 1/2$, meio do intervalo

$F_2 = g(\underbrace{x(k) + q_{21} F_1 \text{ temos}}_{x(k+\frac{1}{2}) = x(t) + F_1 \cdot \frac{\Delta t}{2} \Rightarrow q_{21} = 1/2}, u + \overset{\nearrow}{p_2}) \Rightarrow p_2 = q_{21} = 1/2$

$p_3 = 1$, fim do intervalo

$F_3 = g(\underbrace{x(k) + q_{31} F_1 \Delta t + \overset{0, \text{ uso apenas } F_1}{q_{32}} F_2 \Delta t}_{x(k+1) \Rightarrow x(k) + F_1 \Delta t \Rightarrow q_{31} = 1}, u + \overset{\nearrow}{p_3}) \Rightarrow p_3 = 1, q_{31} = 1, q_{32} = 0$

2) a) $SP(s) = 5 \cdot k \cdot u(t)$

$Y(t) = \underset{\theta_1}{5} + (\underset{\theta_2}{175.5 + 400.7j}) \cdot e^{(\overset{\theta_3}{-5+7.7j})} + (\underset{\theta_4}{175.5 - 400.7j}) \cdot e^{(\overset{\theta_5}{-5-7.7j})}$

$G(s) = \frac{Y(s)}{SU(s)} = 1 + \frac{(175 + 400.7)/5 \cdot s}{s+5-7.7j} + \frac{175.5 - 400.7j)k}{s+5+7.7j} = \frac{84,3}{s^2 - 10s +}$

$$\boxed{G(s) = \frac{84,3}{s^2+10s+84,24}}$$

$H(s) = \frac{10}{s^2+10s} \rightarrow$

$Y(s) = H_1H_2 \cdot (X(t) - Y(t))$

$Y(t) = H_1H_2\, x(t) + H_1H_2\, Y(t)$

$Y(t)(1 + H_1H_2) =$

$Y(t) = \frac{H_1H_2\, sp(s)}{1 + H_1H_2 s}$

$A = \begin{bmatrix} 0 & 1 \\ -84,3 & -10 \end{bmatrix}$

$B = \begin{bmatrix} 0 \\ 84,3 \end{bmatrix}$

$C = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

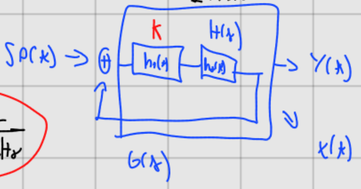
$D = 0$

$A \begin{bmatrix} 0 & 1 \\ 0 & -10 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 10 \end{bmatrix}$

$\frac{Y(s)}{X(s)} = G(s)$

$$\boxed{G(s) = \frac{H_1H_2}{1+H_1H_2}}$$



$x(t) \; \boxed{\frac{h_1(s) \cdot h_2(s)}{1 - h_1 h_2(s)}}$

$G(s) = \frac{k \cdot H(s)}{(1 + k \cdot H(s)} = \frac{\frac{k \cdot 10}{\cancel{s^2+10s}}}{\underbrace{1 - \frac{k \cdot 10}{s^2+10s}}_{\frac{s^2+10s-10k}{\cancel{s^2+10s}}}} = \frac{10k}{s^2+10s+10k} \nearrow \boxed{k = 8,4}$

$H(s) = \frac{10}{(s+0)(s+10)} \rightarrow$ polos não estão na parte negativa

$G(s) = \frac{84,5}{s+(-5\ldots)(s+-5\ldots)} \rightarrow$ polos em $-5 \pm 7,7j$

$\boxed{\text{estável}}$

2025.1

$H(s) = \frac{a}{s+a}$

$C(s) = \frac{k}{(s+12)}$

$G(s) = \frac{C(s) \cdot H(s)}{1 + C(s) \cdot H(s)}$

$C(s) \cdot H(s) = \frac{ak}{\cancel{(s+12)(s+a)}}$

$\frac{C(s) \cdot H(s)}{1 + C(s) \cdot H(s)} = \frac{ak}{(s+12)(s+a)+ak}$

$1 + C(s) \cdot H(s) = \frac{ak + (s+12)(s+a)}{\cancel{(s+12)(s+a)}}$

$G(s) = \frac{ak}{s^2 + s(12+a) + 12a + ak}$

$\boxed{\text{se } a = 2 \\ k = 7.6}$

$y/\, C(s) = \frac{k}{12}$

$H(s) = \frac{\theta_1 \theta_3 \theta_5}{(s-\theta_3)(s-\theta_5)} \cdot \frac{1}{5} \swarrow$ grau

$G(s) = \frac{ak}{s^2 + sa + ak}$

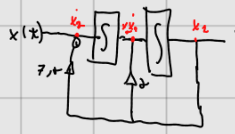
$y + 2\int y + \iint 7{,}2y = \iint 7{,}2u$

$G(s) = \frac{36,17/5}{(s-(-11,23))(s-(-2,85))} = \boxed{\frac{7,27}{s^2 + 14,08s + 32}}$

$G(s) = \frac{7,2}{s^2+2s+7,2}$

$\boxed{y'' + 2y' + 7{,}2y = 7{,}2u}$

$\boxed{H(s) = \frac{2}{s+2}}$



$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$x_1 = Y$

$x_2 = \dot{Y}$

$\dot{x}_2 = \ddot{Y}$

$\dot{x}_2 + 2x_2 + 7{,}2x_1 = 7{,}2u$

$Y = x_2$

$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -7{,}2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 7{,}2 \end{bmatrix} u$

$Y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0]\, u$

2018.1

p/ 3ª ordem sem zeros

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y \\ \dot{y} \\ \ddot{y} \end{bmatrix} \qquad \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -145,4 & -66,14 & -9,445 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 145,4 \end{bmatrix} u$$

$$\frac{-\theta_2\theta_4\theta_6}{(s-\theta_2)(s-\theta_4)(s-\theta_6)}$$

$$= \frac{145,4}{s^3 + 9,44 s^2 + 66,14 s + 145,4} = \frac{132}{}$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \end{bmatrix} u$$

2014.1

$$\frac{\theta_1\theta_3\theta_5}{(s-\theta_3)(s-\theta_5)} = \frac{\frac{180}{12}\cdot(-26)(-42)}{(s+26)(s+42)} = \frac{16380}{s^2+68s+1092} \to y'' + 68y' + 1092y = u\,16380$$

sem zeros ⇒ solução trivial

1.

$$x_1 = y \qquad x_2 = \dot{y} \qquad \dot{x}_1 = x_2$$

$$\dot{x}_2 + 68x_2 + 1092x_1 = 16380u \Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1092 & -68 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 16380 \end{bmatrix} u$$

$$y = x_1$$

se o período for 1 e $H = 0,5s$

$\frac{t}{T}\,\% = 0$

$H > 50\%$ de T

$\frac{t}{2T}\,\% = 0$

enquanto o resto for < 0.5, $u=1$

$> 0,5$, $u=0$

2020.1

1) $y(t) = 1.6 - 1.6e^{-0,3}$

$$H(s) = \frac{-\theta_2\theta_3}{(s-\theta_3)} = \frac{0,48}{(s-0,3)} = \boxed{\frac{0,096}{(s+0,3)}} \to y = x_1$$

$\to \frac{0,096}{j\omega+0,3} \Rightarrow$ p/ $j\omega < 0,3 \Rightarrow$ fica igual; p/ $j\omega > 0,3 \Rightarrow$ dim } para baixo, n tem zeros

$$\frac{0,32}{s} - \frac{0,32}{s+0,3}$$

$$0,32 - \frac{0,32s}{(s+0,3)}$$

$$\frac{\cancel{0,32s} - \cancel{0,32s} + 0,32\cdot 0,3}{s+0,3}$$

$$\dot{y} = \dot{x}_1$$

$$\dot{x}_1 + 0,3x_1 = 0,096u$$

$$\dot{x}_1 = -0,3x_1 + 0,096u$$

$$y = x_1$$

$$A = [-0,3] \qquad B = 0,096$$

$$C = 1 \qquad D = 0$$

$$\cancel{V(s) = sL\cdot I(s) + R\cdot I(s)}$$

$$\cancel{\frac{1}{V(s)} = \frac{1}{sL\cdot I(s) + R\cdot I(s)}}$$

$$\cancel{\frac{I(s)}{V(s)} = \frac{I(s)}{sL\cdot I(s) + R\cdot I(s)}}$$

podia

$$L = 1/0,1 \qquad R = 0,3/0,1$$

$$\frac{0,096}{s+0,3} = \frac{Y(s)}{V(s)} \to s\cdot Y(s) + 0,3Y(s) = 0,096V(s)$$

$$\dot{y}(t) + 0,3y(t) = 0,096$$

2020.1

$$G(s) = \frac{C(s)\cdot H(s)}{1 + C(s)\cdot H(s)}$$

$$\frac{K}{s}\cdot\frac{a}{(s^2+bs+a)} = \frac{Ka}{s^3+bs^2+as}$$